Perturbation Theory

Quantum Physics, 5th Exercise Sheet with Solutions

Exercises

Exercise 1: Spin in a transverse magnetic field (nondegenerate)

Consider

$$H = H^0 + H', \qquad H^0 = -\frac{\hbar\omega_0}{2}\sigma_z, \qquad H' = -\frac{\hbar\omega_1}{2}\sigma_x,$$

with $\omega_1 \ll \omega_0$.

a) Find unperturbed eigenstates and eigenvalues.

Hint: H^0 is proportional to σ_z, so it is already diagonal in the eigenbasis of σ_z.

Solution. The eigenstates of σ_z,

$$\sigma_z |\uparrow\rangle = +|\uparrow\rangle, \qquad \sigma_z |\downarrow\rangle = -|\downarrow\rangle,$$

give

$$H^0 |\uparrow\rangle = -\frac{\hbar\omega_0}{2}|\uparrow\rangle, \qquad H^0 |\downarrow\rangle = +\frac{\hbar\omega_0}{2}|\downarrow\rangle.$$

So $E^{(0)}_\uparrow = -\frac{\hbar\omega_0}{2}$ and $E^{(0)}_\downarrow = +\frac{\hbar\omega_0}{2}$. □

b) Calculate the first- and second-order energy corrections.

Hint: Use $E^{(1)}_n = \langle n| H' |n\rangle$ and $E^{(2)}_n = \sum_{m\neq n} \frac{|\langle m|H'|n\rangle|^2}{E^{(0)}_n - E^{(0)}_m}$. Recall that σ_x flips the spin: $\sigma_x|\uparrow\rangle = |\downarrow\rangle$ and $\sigma_x|\downarrow\rangle = |\uparrow\rangle$.

Solution. First-order energies

$$E^{(1)}_j = \langle j| H' |j\rangle = -\frac{\hbar\omega_1}{2}\langle j|\sigma_x|j\rangle = 0,$$

because σ_x has zero diagonal in the σ_z basis.

Second-order correction for $|\uparrow\rangle$ (only $|\downarrow\rangle$ contributes):

$$E^{(2)}_\uparrow = \sum_{m\neq\uparrow} \frac{|\langle m| H' |\uparrow\rangle|^2}{E^{(0)}_\uparrow - E^{(0)}_m} = \frac{|\langle\downarrow|(-\frac{\hbar\omega_1}{2}\sigma_x)|\uparrow\rangle|^2}{-\hbar\omega_0} = \frac{(\hbar\omega_1/2)^2}{-\hbar\omega_0}.$$

$$E^{(2)}_\downarrow = \sum_{m\neq\downarrow} \frac{|\langle m| H' |\downarrow\rangle|^2}{E^{(0)}_\downarrow - E^{(0)}_m} = \frac{|\langle\uparrow|(-\frac{\hbar\omega_1}{2}\sigma_x)|\downarrow\rangle|^2}{\hbar\omega_0} = \frac{(\hbar\omega_1/2)^2}{\hbar\omega_0}.$$

Hence

$$\boxed{E^{(2)}_\uparrow = -\frac{\hbar\,\omega_1^2}{4\omega_0}}, \qquad \boxed{E^{(2)}_\downarrow = +\frac{\hbar\,\omega_1^2}{4\omega_0}}.$$

(Signs reflect level repulsion: the lower level is pushed down, the upper up.) □

c) Calculate the first-order correction to the eigenstates.

Hint: Use $\left|n^{(1)}\right\rangle = \sum_{m\neq n} \frac{\langle m|H'|n\rangle}{E^{(0)}_n - E^{(0)}_m}|m\rangle$. In a two-level system the sum contains a single term.

Solution. The standard formula

$$\left|\psi^{(1)}_\uparrow\right\rangle = \sum_{m\neq\uparrow} \frac{\langle m| H' |\uparrow\rangle}{E^{(0)}_\uparrow - E^{(0)}_m}|m\rangle$$

yields (single term),

$$|\psi_\uparrow\rangle \approx |\uparrow\rangle + \frac{\langle\downarrow|(-\frac{\hbar\omega_1}{2}\sigma_x)|\uparrow\rangle}{-\hbar\omega_0}|\downarrow\rangle = |\uparrow\rangle + \frac{\omega_1}{2\omega_0}|\downarrow\rangle.$$

Similarly for $|\downarrow\rangle$ with sign reversed. □

d) Compare to exact eigenenergies with the perturbative results: expand the exact energies as a Taylor series in the parameter $\nu = \omega_1/\omega_0$.

Hint: Write $H = -\frac{\hbar}{2}(\omega_0\sigma_z + \omega_1\sigma_x)$ *and use that for* $(\mathbf{v}\cdot\boldsymbol{\sigma})$ *the eigenvalues are* $E_\pm = \pm|\mathbf{v}|$ *for any real vector* $\mathbf{v}$. *For the expansion, use the Taylor series:*

$$f(x+\delta x) = f(x) + f'(x)\delta x + \tag{1}$$

Solution. Exact eigenvalues of the 2×2 Hamiltonian are

$$E_\pm = \pm\frac{\hbar}{2}\sqrt{\omega_0^2+\omega_1^2} = \pm\frac{\hbar\omega_0}{2}\sqrt{1+\nu^2} = \pm\frac{\hbar\omega_0}{2}\left(1+\frac{\nu^2}{2}\right) + \mathcal{O}(\nu^4) = \pm\frac{\hbar\omega_0}{2} \pm \frac{\hbar\omega_1^2}{4\omega_0} + \mathcal{O}(\nu^4),$$

which matches the perturbative results. The perturbative expansion is valid for $\nu \ll 1$; the state-mixing amplitude is $\mathcal{O}(\nu)$. □

Exercise 2: Quantum pendulum as anharmonic oscillator

The energy of a classical pendulum reads as

$$E = \frac{1}{2}mL^2\dot{\phi}^2 + mgL(1-\cos\phi),$$

where L is the length of the pendulum. If we introduce the moment of inertia as $\Theta = mL^2$ and the angular momentum as $N = \Theta\dot{\phi}$, the energy of the pendulum can be written as

$$E = \frac{N^2}{2\Theta} + \Theta\omega_0^2(1-\cos\phi),$$

where we also introduced $\omega_0 = \sqrt{g/L}$. In classical physics, we have learnt that for small angles the pendulum behaves as a harmonic oscillator. Indeed, if we expand $\cos\phi \approx 1 - \frac{\phi^2}{2}$ in leading order, then we get the Hamiltonian

$$H^0 = \frac{N^2}{2\Theta} + \frac{1}{2}\Theta\omega_0^2\phi^2.$$

The quantum pendulum is the quantized version of the pendulum, i.e., N and ϕ are treated as operators similarly to p and x in the harmonic oscillator case. So we can introduce the ladder operator as

$$a = \frac{1}{\sqrt{2}\phi_0}\left(\phi + \frac{i}{\Theta\omega_0}N\right), \quad \phi_0 = \sqrt{\frac{\hbar}{\Theta\omega_0}}.$$

Then the unperturbed Hamiltonian is just the harmonic oscillator

$$H^0 = \hbar\omega_0(a^\dagger a + I/2).$$

Let us expand the cosine further, and treat it as a perturbation:

$$H = H^0 + H', \quad H' = -\frac{\Theta\omega_0^2}{24}\phi^4.$$

a) Calculate the first order corrections to the eigenenergies using the ladder operators:

$$\phi^4 = \frac{\phi_0^4}{4}(a+a^\dagger)^4.$$

Hint: Only the diagonal elements $\langle n|\,(a+a^\dagger)^4\,|n\rangle$ *are needed. Of the 16 operator strings in* $(a+a^\dagger)^4$, *only those with equally many* a*'s and* $a^\dagger$*'s have nonzero diagonal elements (why?). Evaluate the remaining six strings either with* $[a,a^\dagger] = I$, *or by acting on* $|n\rangle$ *directly with* $a\,|n\rangle = \sqrt{n}\,|n-1\rangle$ *and* $a^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$. *Finally, note that* $\Theta\omega_0^2\phi_0^4 = \hbar\omega_0\phi_0^2$.

Solution. In first order, only the diagonal matrix elements are relevant. One needs to evaluate $\langle n|\,(a+a^\dagger)^4\,|n\rangle$, where only terms with equal numbers of a and $a^\dagger$ survive:

$$\langle n|\,(a+a^\dagger)^4\,|n\rangle = \langle n|\,a^2a^{\dagger 2} + aa^\dagger aa^\dagger + aa^{\dagger 2}a + a^{\dagger 2}a^2 + a^\dagger aa^\dagger a + a^\dagger a^2 a^\dagger\,|n\rangle\,.$$

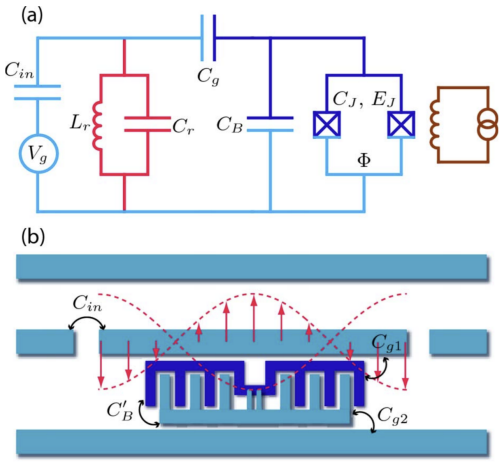


Figure 1: The circuit of a transmon. *Koch et. al, Charge-insensitive qubit design derived from the Cooper pair box, Phys. Rev. A 76, 042319 (2007).*

Let us use the commutator $[a, a^\dagger] = I$ to calculate all terms one-by-one, starting from the end:

$$\begin{aligned}
a^\dagger a^2 a^\dagger &= n[a, a^\dagger] + n^2 = n + n^2, \\
a^\dagger a a^\dagger a &= n^2, \\
a^{\dagger 2} a^2 &= a^\dagger [a^\dagger, a] a + n^2 = n^2 - n, \\
a a^{\dagger 2} a &= [a, a^\dagger] a^\dagger a + n^2 = n^2 + n, \\
a a^\dagger a a^\dagger &= a a^\dagger [a, a^\dagger] + a a^{\dagger 2} a = [a, a^\dagger] + n + n^2 + n = n^2 + 2n + I, \\
a^2 a^{\dagger 2} &= a[a, a^\dagger] a^\dagger + a a^\dagger a a^\dagger = [a, a^\dagger] + n + n^2 + 2n + I = n^2 + 3n + 2I.
\end{aligned}$$

Adding up all the terms, the result is

$$\langle n| (a + a^\dagger)^4 |n\rangle = 3 + 6n + 6n^2.$$

Therefore

$$\boxed{E_n^{(1)} = -\frac{\Theta\omega_0^2}{24} \langle n| \phi^4 |n\rangle = -\frac{\Theta\omega_0^2\phi_0^4}{96}\left(3 + 6n + 6n^2\right) = -\frac{\hbar\omega_0\phi_0^2}{96}\left(3 + 6n + 6n^2\right)}.$$

□

b) Calculate the condition $\langle n| H' |m\rangle \ll \hbar\omega_0$, for which the perturbative approximation of the pendulum is justified. Assume that n, m are small integers.

Hint: Write $\langle n| H' |m\rangle$ as a dimensionless number depending on n and m times an energy scale, and express this energy scale in units of $\hbar\omega_0$ using the definition of ϕ_0. Physically, ϕ_0 is the zero-point spread of the angle.

Solution. The matrix element $\langle n| H' |m\rangle$ equals to some number, which depends on n and m multiplied by $\Theta\omega_0^2\phi_0^4$. If n and m are small integers, let's say for concreteness, less than 3, then the prefactor is less than 1. Then the $\langle n| H' |m\rangle \ll \hbar\omega_0$ condition is equivalent to the $\phi_0^2 \ll 1$:

$$\boxed{\phi_0^2 = \frac{\hbar}{mL^2}\sqrt{\frac{L}{g}} \ll 1}$$

□

Exercise 3: The transmon as a quantum pendulum

Interestingly, one of the most widely used superconducting qubit platforms, the *transmon*, is mathematically equivalent to the *quantum pendulum*. Its simplified circuit is shown in Fig. 1. The Hamiltonian of the transmon reads

$$H_{\text{transmon}} = 4E_C n^2 + E_J(1 - \cos\phi),$$

where n denotes the number of excess Cooper pairs on the superconducting island (blue island in Fig. 1), E_C is the *charging energy* that quantifies the energy cost of adding one Cooper pair, and E_J is the *Josephson energy* associated with the Josephson junction connecting the island to ground.

Upon quantization, the variables n and ϕ become conjugate operators obeying the commutation relation

$$[n, \phi] = -i,$$

which is analogous to the canonical relation between momentum and position. In this sense, the first term $4E_C n^2$ acts as a *kinetic energy*, while the second term $E_J(1 - \cos\phi)$ represents a *potential energy*. The correspondence with the quantum pendulum is therefore

$$n \sim \frac{N}{\hbar}, \qquad \phi \sim \phi, \qquad E_J \leftrightarrow mgL, \qquad \frac{4E_C}{\hbar^2} \leftrightarrow \frac{1}{2\Theta} = \frac{1}{2mL^2}.$$

This mapping shows that the transmon behaves like a quantum pendulum.

a) Approximate the transmon as a harmonic oscillator and calculate its eigenfrequency ω_0 and ϕ_0, which describe the phase uncertainty at the ground state.

Hint: Read off Θ from $\frac{4E_C}{\hbar^2} \leftrightarrow \frac{1}{2\Theta}$, and use $mgL = \Theta\omega_0^2 \leftrightarrow E_J$. Then substitute into $\phi_0 = \sqrt{\hbar/(\Theta\omega_0)}$ from Exercise 2.

Solution.

$$\omega_0 = \sqrt{\frac{g}{L}} = \sqrt{\frac{mgL}{mL^2}} = \sqrt{\frac{8E_C E_J}{\hbar^2}}$$

$$\Theta\omega_0 = \frac{\hbar^2}{8E_C}\sqrt{\frac{8E_C E_J}{\hbar^2}} = \sqrt{\frac{\hbar^2 E_J}{8E_C}}$$

$$\phi_0 = \sqrt{\frac{\hbar}{\Theta\omega_0}} = \sqrt[4]{\frac{8E_C}{E_J}}$$

□

b) What is the condition of treating the transmon as an anharmonic oscillator?

Hint: Translate the condition found in Exercise 2(b) using the result of part (a).

Solution.

$$\phi_0^2 = \sqrt{\frac{8E_C}{E_J}} \ll 1 \iff \frac{E_C}{E_J} \ll 1.$$

□

c) Assume that $E_J/h = 20$ GHz, $E_C/h = 0.3$ GHz. What is the energy difference between the ground and the first excited state, and the energy difference between the first excited and second excited state in the anharmonic oscillator approximation ($\cos\phi \approx 1 - \phi^2/2 + \phi^4/24$), up to first order? What is the anharmonicity $\alpha = E_{01} - E_{12}$?

Hint: Combine $E_n^{(0)} = \hbar\omega_0(n + \frac{1}{2})$ with the first-order correction of Exercise 2(a). Show first that $\hbar\omega_0\phi_0^2 = 8E_C$, so the correction depends on E_C only. Work in frequency units, E/h.

Solution.

$$\frac{\omega_0}{2\pi} = \sqrt{\frac{8E_C E_J}{h^2}} \approx 6.93 \text{ GHz}, \quad \phi_0^2 = \sqrt{\frac{8E_C}{E_J}} \approx 0.35, \quad \hbar\omega_0\phi_0^2 = 8E_C.$$

Using Exercise 2(a),

$$E_n = \hbar\omega_0\Big(n + \frac{1}{2}\Big) - \frac{\hbar\omega_0\phi_0^2}{96}\big(3 + 6n + 6n^2\big) = \hbar\omega_0\Big(n + \frac{1}{2}\Big) - \frac{E_C}{12}\big(3 + 6n + 6n^2\big).$$

This implies that

$$E_0/h = \frac{\omega_0}{4\pi} - \frac{3E_C}{12h} \approx 3.39 \text{ GHz}, \quad E_1/h = \frac{3\omega_0}{4\pi} - \frac{15E_C}{12h} \approx 10.02 \text{ GHz},$$

$$E_2/h = \frac{5\omega_0}{4\pi} - \frac{39E_C}{12h} \approx 16.35 \text{ GHz}.$$

Therefore, the energy differences and the anharmonicity are:

$$\Delta E_{01}/h \approx 6.63 \text{ GHz}, \quad \Delta E_{12}/h \approx 6.33 \text{ GHz}, \quad \alpha/h = \Delta E_{01}/h - \Delta E_{12}/h = 0.30 \text{ GHz}.$$

In first order the anharmonicity is exactly $\alpha = E_C$: the quartic term lowers higher levels more strongly, so the level spacing decreases up the ladder. □

Exercise 4: The nitrogen-vacancy center in transverse magnetic field

The negatively charged nitrogen-vacancy (NV^-) center in diamond consists of a substitutional nitrogen atom adjacent to a vacant lattice site. The static Hamiltonian of the NV^- center ground state in zero magnetic field, neglecting any interactions with nuclear spins or spin-strain interactions, and assuming that the NV^--axis is oriented along the z-direction, is given by

$$H = hDS_z^2, \tag{2}$$

where h is the Planck constant, $D = 2.87$ GHz is the zero-field splitting, and

$$S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \tag{3}$$

is the z component of the dimensionless spin-1 operator. The resulting Hamiltonian has two energy levels: one at zero energy and a doubly degenerate energy level with energy hD, where $D = 2.87$ GHz. The system couples to the magnetic field as

$$h\gamma_e \mathbf{S} \cdot \mathbf{B}, \tag{4}$$

where $\gamma_e = 28$ GHz/T is the electron gyromagnetic ratio, and $\mathbf{S}$ is the vector of the dimensionless spin-1 operators. The x and y components read as

$$S_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad S_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}. \tag{5}$$

a) Show that the spin-1 operators follow the same commutator relations as the spin-1/2 operators $\sigma_i/2$:

$$[S_x, S_y] = iS_z, \quad [S_y, S_z] = iS_x, \quad [S_z, S_x] = iS_y. \tag{6}$$

(These commutation relations define the Lie algebra $\mathfrak{su}(2)$; the spin-1/2 operators $\sigma_i/2$ and the spin-1 matrices above are two different representations of it. All spin and angular momentum operators follow the same commutation relations.)

Hint: Since the S_i are Hermitian, the reversed products follow by conjugation, e.g. $S_yS_x = (S_xS_y)^\dagger$, so only three matrix products need to be computed. Multiplying by the diagonal S_z from the left (right) simply rescales the rows (columns).

Solution.

$$S_xS_y - S_yS_x = \frac{1}{2}\begin{pmatrix} i & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & -i \end{pmatrix} - \frac{1}{2}\begin{pmatrix} -i & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & i \end{pmatrix} = iS_z$$

$$S_yS_z - S_zS_y = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 0 & 0 \\ i & 0 & i \\ 0 & 0 & 0 \end{pmatrix} - \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & -i & 0 \\ 0 & 0 & 0 \\ 0 & -i & 0 \end{pmatrix} = iS_x$$

$$S_zS_x - S_xS_z = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} - \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} = iS_y$$

□

b) Calculate the eigenvalues and eigenvectors of $H^0 = hDS_z^2$.

Hint: H^0 is diagonal: $S_z^2 = \mathrm{diag}(1, 0, 1)$.

Solution.

$$E_0 = 0, \quad |\phi_0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad E_1 = hD, \quad |\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad E_2 = hD, \quad |\phi_2\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

□

c) Let us apply a small, static magnetic field in the x direction: $H' = h\gamma_e B S_x$. Calculate the first- and second-order energy correction of the ground state.

Hint: Since $|\phi_0\rangle$ is the middle basis vector, the matrix elements $\langle\phi_m| S_x |\phi_0\rangle$ are simply the entries of the middle column of S_x. Both excited states have the same energy denominator.

Solution.

$$E_0^{(1)} = \langle\phi_0| H' |\phi_0\rangle = 0,$$
$$E_0^{(2)} = \sum_{m\neq 0} \frac{|\langle\phi_m| H' |\phi_0\rangle|^2}{E_0^{(0)} - E_m^{(0)}} = -\frac{h\gamma_e^2 B^2}{D}$$

□

d) Calculate the first-order correction to the ground state.

Hint: Use the same matrix elements as in (c), together with $\left|n^{(1)}\right\rangle = \sum_{m\neq n} \frac{\langle\phi_m|H'|\phi_n\rangle}{E_n^{(0)} - E_m^{(0)}} |\phi_m\rangle$.

Solution.

$$\left|\phi_0^{(1)}\right\rangle = \sum_{m\neq 0} \frac{\langle\phi_m| H' |\phi_0\rangle}{E_0^{(0)} - E_m^{(0)}} |\phi_m\rangle = -\frac{\gamma_e B}{\sqrt{2}D}(|\phi_1\rangle + |\phi_2\rangle) = -\frac{\gamma_e B}{\sqrt{2}D}\begin{pmatrix}1\\0\\1\end{pmatrix}.$$

□

e) Use quasi-degenerate perturbation theory, to derive an effective Hamiltonian for the two excited states.

Hint: The degenerate subspace is spanned by $|\phi_1\rangle, |\phi_2\rangle$. *First check whether* H' *couples these two states directly. In second order the only intermediate state outside the subspace is* $|\phi_0\rangle$, *so the sum in* $H^{(2)}$ *has a single term.*

Solution. Let us use the formulas from the lecture:

$$H^{(1)}_{mm'} = H'_{mm'} = \langle\phi_m| H' |\phi_{m'}\rangle \implies H^{(1)} = 0$$

Second-order:

$$H^{(2)}_{mm'} = \frac{1}{2}\langle\phi_m| H' |\phi_0\rangle \langle\phi_0| H' |\phi_{m'}\rangle \left[\frac{1}{E_m - E_0} + \frac{1}{E_{m'} - E_0}\right] \implies H^{(2)} = \frac{h\gamma_e^2 B^2}{2D}\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}.$$

So the effective Hamiltonian up to second-order reads as

$$\tilde{H} = h\begin{pmatrix}D + \frac{\gamma_e^2 B^2}{2D} & \frac{\gamma_e^2 B^2}{2D}\\ \frac{\gamma_e^2 B^2}{2D} & D + \frac{\gamma_e^2 B^2}{2D}\end{pmatrix} = h\left(D + \frac{\gamma_e^2 B^2}{2D}\right)I + \frac{h\gamma_e^2 B^2}{2D}\sigma_x$$

□

f) Calculate the eigenstates and eigenenergies of the effective Hamiltonian.

Hint: A matrix of the form $aI + b\sigma_x$ *shares its eigenvectors with* σ_x, *and its eigenvalues are* $a \pm b$.

Solution. The eigenstate-eigenenergy pairs of the effective Hamiltonian read as

$$\tilde{E}_+ = hD + \frac{h\gamma_e^2 B^2}{D} \;\to\; |+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix},$$

$$\tilde{E}_- = hD \;\to\; |-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$$

□

g) Calculate the generator of the unitary transformation S up to first order.

Hint: $S^{(1)}$ *is anti-Hermitian,* $S^{(1)\dagger} = -S^{(1)}$, *and in first order it only connects the degenerate subspace with* $|\phi_0\rangle$, *with elements of magnitude* $|\langle\phi_m| H' |\phi_0\rangle|/|E_m - E_0|$. *The sign is fixed by requiring that* $(I + S^{(1)})|\phi_0\rangle$ *reproduces the corrected ground state of (d).*

Solution.

$$S^{(1)}_{m0} = -\frac{\langle\phi_m| H' |\phi_0\rangle}{hD} \implies S^{(1)} = -\frac{\gamma_e B}{\sqrt{2}D}(|\phi_1\rangle\langle\phi_0| + |\phi_2\rangle\langle\phi_0| - h.c.) = \frac{\gamma_e B}{\sqrt{2}D}\begin{pmatrix}0 & -1 & 0\\1 & 0 & 1\\0 & -1 & 0\end{pmatrix}$$

□

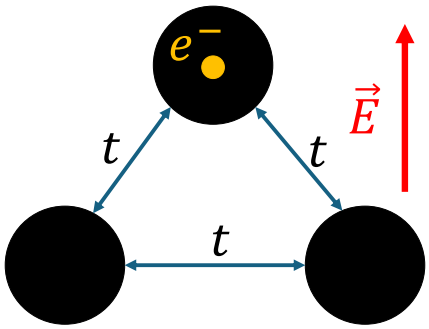


Figure 2: Three-site, symmetric tight-binding model in finite electric field.

h) Express the corrected eigenstates and eigenenergies of the excited states up to first order: first, express the solutions of the effective Hamiltonian as states of the original model, then rotate these states with the unitary $e^S \approx I + S^{(1)}$.

Hint: The components of $|\pm\rangle$ are the coefficients of $|\phi_1\rangle$ and $|\phi_2\rangle$. Before multiplying with $S^{(1)}$, check how S_x acts on the two combinations: one of them is completely decoupled from $|\phi_0\rangle$.

Solution.

$$\left|\tilde{\phi}_1\right\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle - |\phi_2\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\-1\end{pmatrix}, \quad \left|\tilde{\phi}_2\right\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle + |\phi_2\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\1\end{pmatrix}$$

$$\left|\phi_1^{\text{pert}}\right\rangle = \left|\tilde{\phi}_1\right\rangle + S^{(1)}\left|\tilde{\phi}_1\right\rangle = \left|\tilde{\phi}_1\right\rangle, \quad \left|\phi_2^{\text{pert}}\right\rangle = \left|\tilde{\phi}_2\right\rangle + S^{(1)}\left|\tilde{\phi}_2\right\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\1\end{pmatrix} + \frac{\gamma_e B}{D}\begin{pmatrix}0\\1\\0\end{pmatrix}$$

□

i) Check the orthonormality of the perturbed eigenstates.

Hint: Write all three states in the basis $|\phi_0\rangle, |\phi_1\rangle, |\phi_2\rangle$. The perturbative states only need to be orthonormal up to the order of the calculation, so deviations of order $(\gamma_e B/D)^2$ are acceptable.

Solution. The perturbed eigenstates can be expressed using the original orthonormal basis:

$$\left|\phi_0^{\text{pert}}\right\rangle = |\phi_0\rangle - \frac{\gamma_e B}{\sqrt{2}D}(|\phi_1\rangle + |\phi_2\rangle), \quad \left|\phi_1^{\text{pert}}\right\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle - |\phi_2\rangle), \quad \left|\phi_2^{\text{pert}}\right\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle + |\phi_2\rangle) + \frac{\gamma_e B}{D}|\phi_0\rangle$$

Then, the scalar products:

$$\left\langle\phi_0^{\text{pert}}\middle|\phi_0^{\text{pert}}\right\rangle = 1 + \frac{\gamma_e^2 B^2}{D^2}, \quad \left\langle\phi_1^{\text{pert}}\middle|\phi_1^{\text{pert}}\right\rangle = 1, \quad \left\langle\phi_2^{\text{pert}}\middle|\phi_2^{\text{pert}}\right\rangle = 1 + \frac{\gamma_e^2 B^2}{D^2},$$

$$\left\langle\phi_0^{\text{pert}}\middle|\phi_1^{\text{pert}}\right\rangle = 0, \quad \left\langle\phi_0^{\text{pert}}\middle|\phi_2^{\text{pert}}\right\rangle = 0, \quad \left\langle\phi_1^{\text{pert}}\middle|\phi_2^{\text{pert}}\right\rangle = 0.$$

The states are orthogonal, and normalized up to second-order terms, which lie beyond the accuracy of the first-order calculation. □

Exercise 5: Three-site triangular tight-binding (degenerate perturbation theory)

The tight-binding model is a simple and widely used description of electrons in solids or artificial lattices. It assumes that the electron can occupy a set of localized sites (such as atoms or quantum dots), and can "hop" between neighboring sites with a certain amplitude t. The quantum state of the electron is then a superposition of being on each site. In this exercise, we consider a minimal example, a symmetric triangular molecule or ring of three sites, where the electron can hop between any pair of sites, see the figure.

The unperturbed Hamiltonian (site basis $|1\rangle, |2\rangle, |3\rangle$):

$$H^0 = -t\begin{pmatrix}0 & 1 & 1\\1 & 0 & 1\\1 & 1 & 0\end{pmatrix}, \qquad t > 0.$$

a) Find the eigenvalue-eigenvector pairs for H^0.

Hint: The problem is symmetric under the permutation of sites, so the eigenstates can also be searched in symmetric forms: the occupation of the sites is uniform for each eigenstate. Use the Ansatz $|\phi\rangle = \frac{1}{\sqrt{3}}\left(1,\, e^{i\alpha},\, e^{i\beta}\right)^T$. *Alternatively, write* $H^0 = -t(J - I)$, *where* J *is the matrix of all ones, whose eigenvalues are easy to guess.*

Solution. The fully symmetric state

$$|\phi_0\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

has eigenvalue

$$E_0 = -2t.$$

The orthogonal subspace is two-dimensional with eigenvalue $E_{1,2} = +t$. One convenient orthonormal basis for the degenerate subspace is

$$|\phi_1\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ e^{i\frac{2\pi}{3}} \\ e^{-i\frac{2\pi}{3}} \end{pmatrix}, \qquad |\phi_2\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ e^{-i\frac{2\pi}{3}} \\ e^{i\frac{2\pi}{3}} \end{pmatrix},$$

both with eigenvalue $+t$. □

b) Let us switch on a finite, but small electric field as in Fig. 2. The potential of an electron in electric field is $V = e\boldsymbol{\mathcal{E}} \cdot \mathbf{r}$. This means that the bottom two sites have the same potential, and the top site has a different, larger potential. If we choose the origin conveniently, then the electric potential is equivalent to the perturbation:

$$H' = \begin{pmatrix} \varepsilon & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Calculate the energy correction to the ground state up to second order.

Hint: Since $H' = \varepsilon\,|1\rangle\langle 1|$, *every matrix element factorizes:* $\langle\phi_m|\,H'\,|\phi_n\rangle = \varepsilon\,\langle\phi_m|1|\phi_m|1\rangle\,\langle 1|\phi_n|1|\phi_n\rangle$, *and every eigenstate has amplitude* $1/\sqrt{3}$ *on site 1.*

Solution. For the symmetric ground state

$$E_0^{(1)} = \langle\phi_0|\,H'\,|\phi_0\rangle = \frac{\varepsilon}{3},$$

$$E_0^{(2)} = \sum_{m\neq 0} \frac{|\,\langle\phi_m|\,H'\,|\phi_0\rangle\,|^2}{E_0^{(0)} - E_m^{(0)}} = \frac{2\varepsilon^2}{-27t}$$

□

c) Calculate the first-order correction to the ground state.

Hint: Use the matrix elements from (b). To interpret the result in the site basis, use $e^{i2\pi/3} + e^{-i2\pi/3} = -1$.

Solution.

$$\left|\phi_0^{(1)}\right\rangle = \sum_{m\neq 0} \frac{\langle\phi_m|\,H'\,|\phi_0\rangle}{E_0^{(0)} - E_m^{(0)}}\,|\phi_m\rangle = -\frac{\varepsilon}{9t}(|\phi_1\rangle + |\phi_2\rangle) = \frac{\varepsilon}{9\sqrt{3}t} \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}.$$

The electric field breaks the symmetry, and pulls the electron to the bottom, as expected. □

d) Use quasi-degenerate perturbation theory, to derive an effective Hamiltonian for the two excited states.

Hint: Unlike in Exercise 4, here H' *couples* $|\phi_1\rangle$ *and* $|\phi_2\rangle$ *already in first order. In second order the only intermediate state is* $|\phi_0\rangle$, *with energy denominator* $E_{1,2} - E_0 = 3t$.

Solution. Let us use the formulas from the lecture:

$$H^{(1)}_{mm'} = H'_{mm'} = \langle\phi_m| H' |\phi_{m'}\rangle \implies H^{(1)} = \frac{\varepsilon}{3}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$$

Second-order:

$$H^{(2)}_{mm'} = \frac{1}{2}\langle\phi_m| H' |\phi_0\rangle \langle\phi_0| H' |\phi_{m'}\rangle \left[\frac{1}{E_m - E_0} + \frac{1}{E_{m'} - E_0}\right] \implies H^{(2)} = \frac{\varepsilon^2}{27t}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}.$$

So the effective Hamiltonian up to second-order reads as

$$\tilde{H} = \begin{pmatrix} t + \frac{\varepsilon}{3} + \frac{\varepsilon^2}{27t} & \frac{\varepsilon}{3} + \frac{\varepsilon^2}{27t} \\ \frac{\varepsilon}{3} + \frac{\varepsilon^2}{27t} & t + \frac{\varepsilon}{3} + \frac{\varepsilon^2}{27t}\end{pmatrix} = \left(t + \frac{\varepsilon}{3} + \frac{\varepsilon^2}{27t}\right) I + \left(\frac{\varepsilon}{3} + \frac{\varepsilon^2}{27t}\right)\sigma_x$$

□

e) Calculate the eigenstates and eigenenergies of the effective Hamiltonian.
Hint: The effective Hamiltonian has the same structure as in Exercise 4(f): $\tilde{H} = aI + b\sigma_x$.

Solution. The eigenstate-eigenenergy pairs of the effective Hamiltonian read as

$$\tilde{E}_+ = t + \frac{2\varepsilon}{3} + \frac{2\varepsilon^2}{27t} \to |+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix},$$

$$\tilde{E}_- = t \to |-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$$

□

f) Calculate the generator of the unitary transformation S up to first order.
Hint: Proceed as in Exercise 4(g), now with energy denominator $3t$. Write the matrix in the eigenbasis $|\phi_0\rangle, |\phi_1\rangle, |\phi_2\rangle$ of H^0, not in the site basis.

Solution. In the basis $(|\phi_0\rangle, |\phi_1\rangle, |\phi_2\rangle)$:

$$S^{(1)}_{m0} = -\frac{\langle\phi_m| H' |\phi_0\rangle}{3t} \implies S^{(1)} = -\frac{\varepsilon}{9t}(|\phi_1\rangle\langle\phi_0| + |\phi_2\rangle\langle\phi_0| - h.c.) = \frac{\varepsilon}{9t}\begin{pmatrix}0 & 1 & 1\\ -1 & 0 & 0\\ -1 & 0 & 0\end{pmatrix}$$

□

g) Express the corrected eigenstates and eigenenergies of the excited states up to first order: first, express the solutions of the effective Hamiltonian as states of the original model, then rotate these states with the unitary e^S.
Hint: Use $e^{i2\pi/3} - e^{-i2\pi/3} = i\sqrt{3}$ and $e^{i2\pi/3} + e^{-i2\pi/3} = -1$ to write $\left|\tilde{\phi}_{1,2}\right\rangle$ in the site basis. Which of them has zero amplitude on site 1, and what does this imply for its coupling to $|\phi_0\rangle$?

Solution.

$$\left|\tilde{\phi}_1\right\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle - |\phi_2\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ i\\ -i\end{pmatrix}, \quad \left|\tilde{\phi}_2\right\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle + |\phi_2\rangle) = \frac{1}{\sqrt{6}}\begin{pmatrix}2\\ -1\\ -1\end{pmatrix}$$

$$e^S \approx I + S^{(1)}, \quad \left|\phi_1^{\text{pert}}\right\rangle = \left|\tilde{\phi}_1\right\rangle + S^{(1)}\left|\tilde{\phi}_1\right\rangle = \left|\tilde{\phi}_1\right\rangle, \quad \left|\phi_2^{\text{pert}}\right\rangle = \left|\tilde{\phi}_2\right\rangle + S^{(1)}\left|\tilde{\phi}_2\right\rangle = \frac{1}{\sqrt{6}}\begin{pmatrix}2\\ -1\\ -1\end{pmatrix} + \frac{\sqrt{2}\varepsilon}{9\sqrt{3}t}\begin{pmatrix}1\\ 1\\ 1\end{pmatrix}$$

□

h) Check the orthonormality of the perturbed eigenstates.
Hint: Work in the orthonormal basis $|\phi_0\rangle, \left|\tilde{\phi}_1\right\rangle, \left|\tilde{\phi}_2\right\rangle$, and compare with Exercise 4(i).

Solution. The perturbed eigenstates can be expressed using the orthonormal basis: $|\phi_0\rangle, \left|\tilde{\phi}_1\right\rangle, \left|\tilde{\phi}_2\right\rangle$:

$$\left|\phi_0^{\text{pert}}\right\rangle = |\phi_0\rangle - \frac{\sqrt{2}\varepsilon}{9t}\left|\tilde{\phi}_2\right\rangle, \quad \left|\phi_1^{\text{pert}}\right\rangle = \left|\tilde{\phi}_1\right\rangle, \quad \left|\phi_2^{\text{pert}}\right\rangle = \left|\tilde{\phi}_2\right\rangle + \frac{\sqrt{2}\varepsilon}{9t}|\phi_0\rangle$$

Then, the scalar products:

$$\left\langle\phi_0^{\text{pert}}\middle|\phi_0^{\text{pert}}\right\rangle = 1 + \frac{2\varepsilon^2}{81t^2}, \quad \left\langle\phi_1^{\text{pert}}\middle|\phi_1^{\text{pert}}\right\rangle = 1, \quad \left\langle\phi_2^{\text{pert}}\middle|\phi_2^{\text{pert}}\right\rangle = 1 + \frac{2\varepsilon^2}{81t^2},$$

$$\left\langle\phi_0^{\text{pert}}\middle|\phi_1^{\text{pert}}\right\rangle = 0, \quad \left\langle\phi_0^{\text{pert}}\middle|\phi_2^{\text{pert}}\right\rangle = 0, \quad \left\langle\phi_1^{\text{pert}}\middle|\phi_2^{\text{pert}}\right\rangle = 0.$$

As in Exercise 4, the states are orthogonal and normalized up to second-order terms. □